



BANK MANAGEMENT SYSTEM

A Java-Based OOP Project

Project Owners:



Rayen Taleb



Ahmed Houcine Slimi

PROJECT OVERVIEW

A comprehensive Java Object-Oriented Programming (OOP) banking management system with both console and graphical interfaces, demonstrating all core OOP concepts.



Technology

Java Programming Language



Architecture

Object-Oriented Design



Core Packages

- Models (src/models/)
- Services (src/services/)



OOP Concepts

- Inheritance
- Composition
- Encapsulation

MODELS PACKAGE STRUCTURE

User.java

Base Class for All Users

- Personal information storage
- Authentication (login/logout)
- Access control methods

Admin.java

Administrator with Elevated Privileges

- Inherits from User
- Enhanced security validation
- Access level management

Account.java

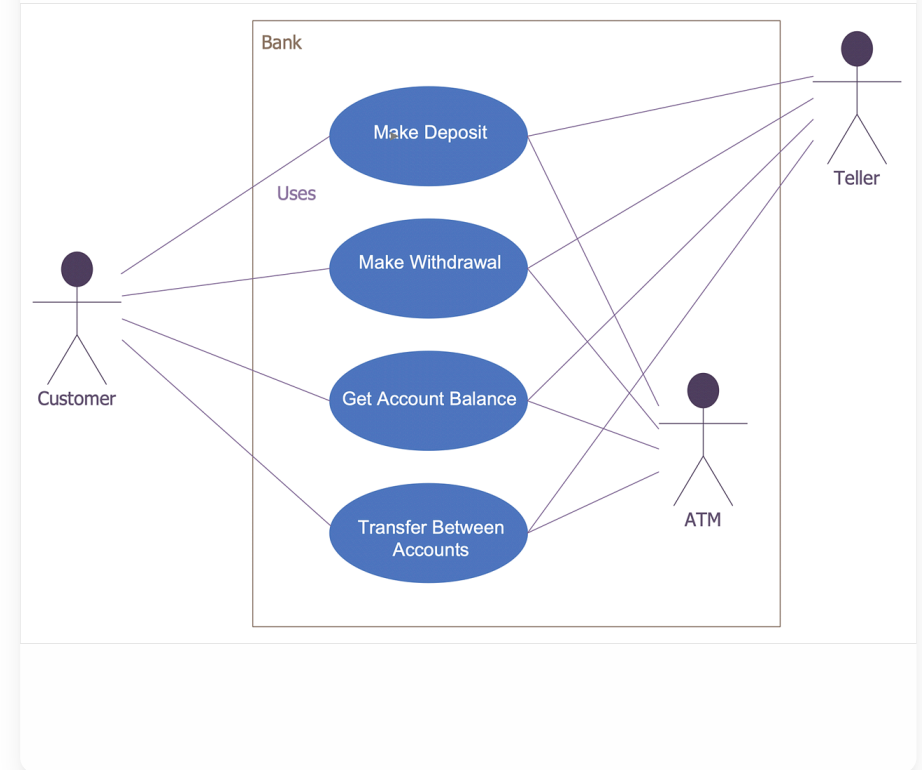
Bank Account with Balance Management

- Balance tracking
- Transaction history
- Links to User object

Operation.java

Transaction Record

- DEPOSIT/WITHDRAW types
- Amount & timestamp
- Balance tracking



Package Architecture Overview

USER & ADMIN CLASSES

User.java

Base Class for All Users

- **Personal Information Storage**

User ID, Username, Password, Name, Email, Phone, Address

- **Authentication System**

login() and logout() functions

- **Access Control**

Getters and Setters for controlled data access

- **Display Functionality**

displayPersonalInfo() method

Admin.java

Administrator with Elevated Privileges

- **Inherits from User**

All personal info fields, login/logout, display methods

- **Admin-Specific Fields**

adminId, department, accessLevel

- **Enhanced Security**

Overridden login() with access level validation

- **Administrative Operations**

Manage accounts, operations, and view all data

ACCOUNT & OPERATION CLASSES

Account.java

Bank Account with Balance Management

- **Account Details**

accountNumber, balance, accountType, status, createdAt

- **Balance Management**

Deposit and withdrawal tracking

- **Transaction History**

ArrayList of all operations

- **User Association**

Links to account holder (User object)

Operation.java

Transaction Record

- **Transaction Details**

operationId, type (DEPOSIT/WITHDRAW), amount

- **Account Reference**

accountNumber, balanceAfter transaction

- **Timestamp & Description**

operationDate (automatic), optional notes

- **Helper Methods**

isDeposit(), isWithdrawal() classification

SERVICES PACKAGE – BANKSERVICE

Data Structures

HashMaps for Fast Lookup

Admin Ops

- Account/Operation Mgmt
- View all data
- Bank statistics
- Add/delete items

User Ops

- Registration & Login
- View personal info
- Modify details
- View own operations

Transactions

- Deposit money
- Withdraw funds
- Auto record creation
- Insufficient funds check

Utilities

- Get by ID methods
- Count operations
- Display statistics

Security

- Admin authentication
- Access level validation
- User credential check

Integration

- Unified interface
- Data integrity
- Business rules enforcement

TESTING & DEMONSTRATION

Test.java

✓ **14 Test Scenarios**
Covers all OOP concepts

 **Interactive Mode**
Keyboard input available

✓ **Comprehensive Coverage**
Validates system behavior

<> Demonstrates OOP

- Inheritance
- Composition
- Encapsulation

Test Scenarios

1 **System Init**
Initialize bank service

2 **Account Mgmt**
Add/delete accounts

3 **User Auth**
Registration & login

4 **Deposit Ops**
Money deposit tests

5 **Withdrawal**
Fund withdrawal tests

6 **User View**
Display functionality

10 **Inheritance**
Admin class demo

12 **Encapsulation**
Access control demo

SYSTEM FEATURES & CAPABILITIES

User Management

- Registration & authentication
- Personal info storage
- Login/logout session control
- Information modification

Account Management

- Multiple account types
- Balance tracking
- Account status control
- Transaction history

Transaction Processing

- Deposit & withdrawal
- Insufficient funds check
- Automatic operation logging
- Balance after transaction

Administrative Control

- Elevated privileges
- Access level validation
- Full data visibility
- Admin-specific operations

OOP Implementation

- Inheritance (Admin extends User)
- Composition (Account has Operations)
- Encapsulation (private fields)
- Access control methods

Data Integrity

- Centralized data management
- HashMap fast lookup
- Business rule enforcement
- Consistent state maintenance

Bank Management System — A comprehensive Java-based solution demonstrating modern OOP principles